Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**AIM: Perform FFT and IFFT on discrete time signal**	
Experiment No: 5	**Date:01-10-2026**	**Enrollment No:92400133121**

Aim: Perform FFT and IFFT on discrete time signal

Code:

```
import numpy as np
import matplotlib.pyplot as plt

signal = np.array([2,4,6,7,9])

fft_result = np.fft.fft(signal)
print(fft_result)

magnitude_spectrum = np.abs(fft_result)
print(magnitude_spectrum)

phase_spectrum = np.angle(fft_result)
print(phase_spectrum)

reconstructed_signal = np.fft.ifft(fft_result)
print(reconstructed_signal)

sample_index = np.arange(len(signal))

plt.figure(figsize=(12,8))

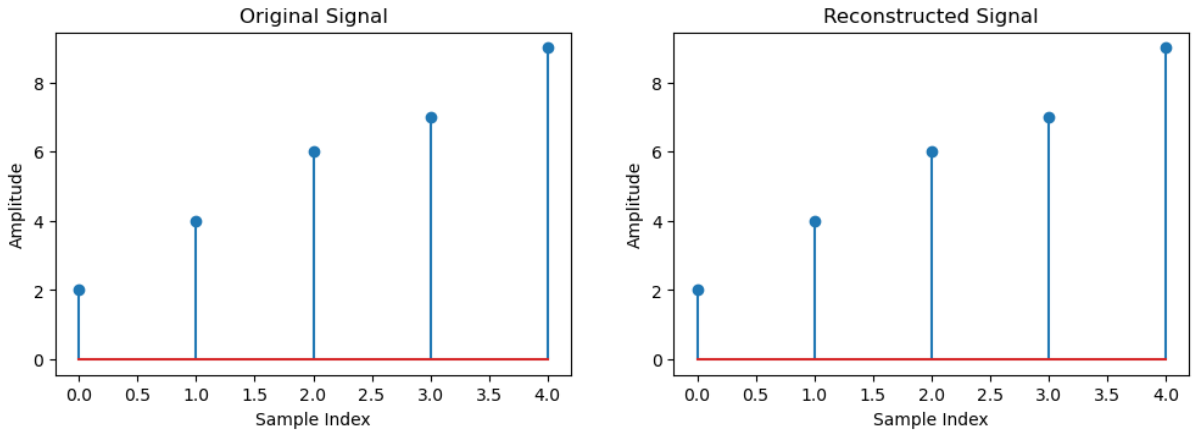
plt.subplot(2,2,1)
plt.stem(sample_index,signal)
plt.title("Original Signal")
plt.xlabel("Sample Index")
plt.ylabel("Amplitude")

plt.subplot(2,2,2)
plt.stem(sample_index,reconstructed_signal)
plt.title("Reconstructed Signal")
plt.xlabel("Sample Index")
plt.ylabel("Amplitude")

plt.show()
```

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Output:



Conclusion:

- In this experiment the Fast Fourier Transform and Inverse Fast Fourier Transform were successfully performed.
- The FFT was used to convert the discrete time domain signal into its frequency domain representation, from which the magnitude and phase spectrum were observed.
- The original time domain signal was perfectly reconstructed.